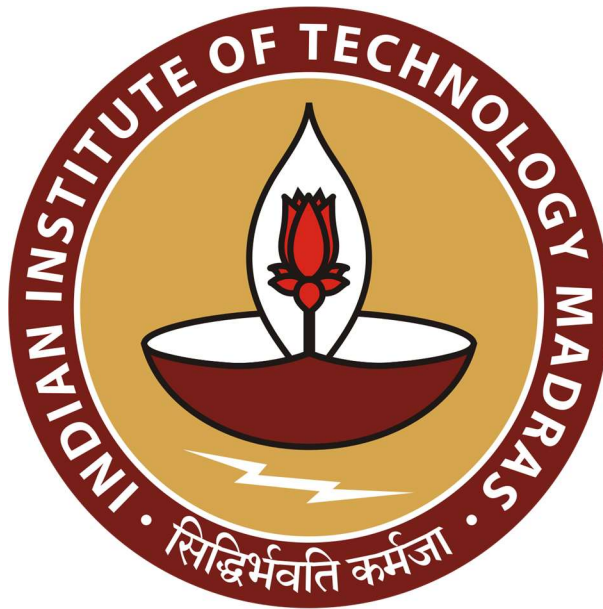


CV5570 – Computer Vision for Construction Engineering

Image Stitching Using Classical and Deep-Learning Based Feature Matching



Assignment 01

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Assignment 1

CV5570 – Computer Vision for Construction Engineering

Objective

The goal of this assignment is to construct and evaluate an image stitching pipeline using two different approaches:

1. A classical computer vision approach based on hand crafted local features and feature matching.
2. A deep learning approach to learned local features and descriptors.

The classical approach was investigated using traditional feature detection and description methods such as SIFT, ORB and SURF. ORB was chosen for the final classical stitching pipeline. The deep learning approach was implemented using R2D2, a learned local feature detector and descriptor.

The main goal is to understand how local features are detected and matched between overlapping images, how the geometrical relationship between the images is estimated using homography, and how the individual images can then be warped and composited so that they produce a panorama.

Both approaches are compared in terms of feature quality, matching behaviour, computational considerations, stitching quality, robustness and failure cases.

Images/Dataset

Three images were taken in front of the ANRF National Facility for Cryo-Electron Microscopy Building, IIT Madras Campus using Samsung Galaxy S24 Ultra.

The pictures were taken in sequence with overlapping areas so that adjacent pictures could be matched and merged into a panorama.

Image 1



Image 2



Image 3



Image Metadata:

Image	Resolution / Quality	ISO	Focal Length	Exposure Compensation	Aperture	Shutter Speed	File Size
Image 1	9 MP	12	23 mm	0.0 EV	F1.7	1/180 s	4.12 MB
Image 2	9 MP	12	23 mm	0.0 EV	F1.7	1/585 s	3.65 MB
Image 3	9 MP	12	23 mm	0.0 EV	F1.7	1/720 s	3.89 MB

The three images have the same nominal resolution, ISO, focal length, aperture and exposure compensation. However, the shutter speed changes between the images because of the change in scene illumination and camera position.

The images contain architectural structures, edges, windows, walls and other visual features that provide suitable local information for feature detection and matching.

Classical Methods

Classical image stitching methods rely on hand-crafted local features to identify corresponding regions between overlapping images. In this assignment, three commonly used feature-based methods, namely SIFT, ORB and SURF, were considered. These methods consist of two main components: keypoint detection and local feature description. The detected features from two images are then matched to identify corresponding image locations.

SIFT

The Scale-Invariant Feature Transform (SIFT) was introduced by Lowe [1]. SIFT detects distinctive keypoints in a scale space constructed using progressively blurred versions of the image. Keypoints are identified using extrema in the Difference-of-Gaussian scale space. An orientation is assigned to each keypoint based on local image gradients, providing robustness to image rotation.

A descriptor is then generated from the local gradient distribution around each keypoint. Since the descriptor captures local appearance rather than raw pixel values, SIFT provides good robustness to moderate changes in scale, rotation and illumination.

In this assignment, SIFT was used to examine feature correspondences between the input images. The resulting matches demonstrate that architectural structures such as corners, edges and other locally distinctive regions can be used to establish correspondences.

The main limitation of SIFT is its relatively high computational cost compared with binary feature descriptors. Its floating-point descriptor also requires more memory and computational effort for matching.

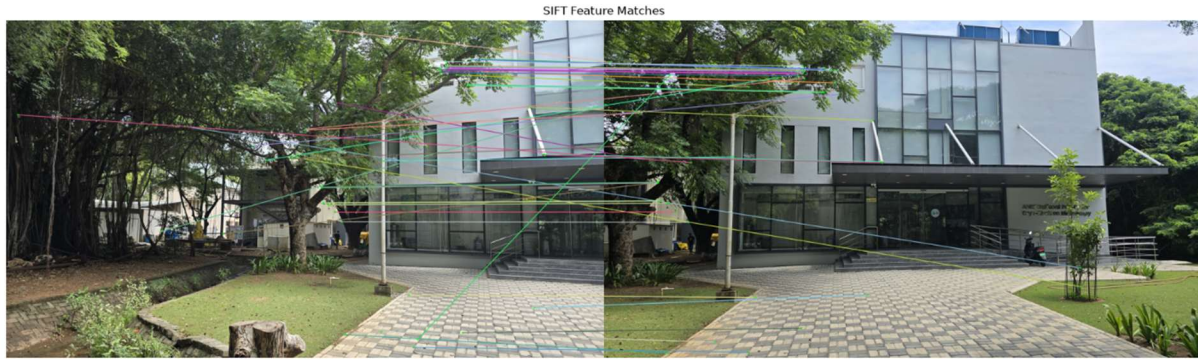
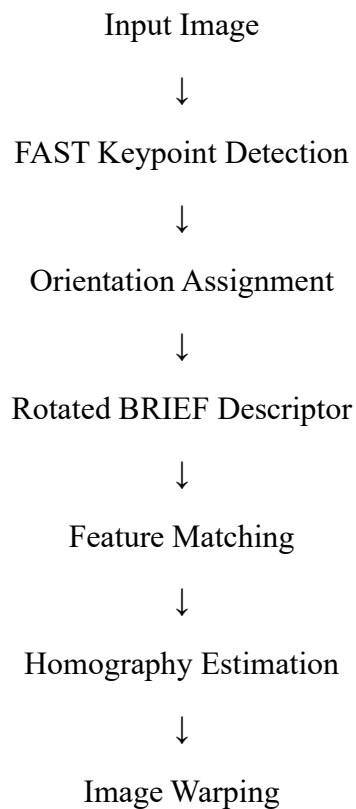


Fig 1. SIFT feature matching between overlapping images

ORB

ORB (Oriented FAST and Rotated BRIEF) was proposed by Rublee et al. [2] as an efficient alternative to SIFT and SURF. ORB combines the FAST keypoint detector with an oriented version of the BRIEF descriptor.

The general ORB pipeline used in this work is:



ORB produces a binary descriptor, allowing efficient descriptor matching using Hamming distance. This makes ORB considerably more computationally efficient than methods based on high-dimensional floating-point descriptors.



Fig 2. Representative ORB feature matches

For the Image 1–Image 2 pair, the implementation produced **105 ORB matches**, of which **80 were geometrically consistent inliers** after homography estimation. The corresponding inlier ratio is:

$$\text{Inlier Ratio} = \frac{80}{105} \times 100 = 76.19\%$$

For Image 2–Image 3, **58 ORB matches** were obtained. The lower number of matches indicates that this pair was more challenging for feature matching than Image 1–Image 2.

ORB was selected as the final classical method because it provides a good balance between matching quality and computational efficiency.

SURF

Speeded-Up Robust Features (SURF) was proposed by Bay et al. [3]. SURF uses a Hessian-matrix-based detector and Haar-wavelet-based descriptors. It was designed to provide scale and rotation robustness while reducing the computational cost associated with SIFT.

SURF generally provides more computationally efficient feature extraction than the original SIFT formulation but still uses floating-point descriptors and is computationally more demanding than ORB.

The three classical methods can therefore be summarized as follows:

Property	SIFT	ORB	SURF
Feature type	Hand-crafted	Hand-crafted	Hand-crafted
Descriptor	Floating point	Binary	Floating point
Scale robustness	High	Moderate	High
Rotation robustness	High	Good	High

Matching	L2-based	Hamming	L2-based
Computational cost	High	Low	Medium
Descriptor size	Large	Compact	Medium
Main advantage	Robustness	Efficiency	Robustness + efficiency

Based on these characteristics and the observed matching performance, ORB was used for the final classical stitching pipeline.

Classical Stitching Implementation

The classical stitching pipeline was implemented sequentially using the three images:

Image 1 ↔ Image 2



Intermediate Panorama



Image 3



Final Panorama

For each pair of overlapping images, ORB keypoints and descriptors were extracted first. Corresponding descriptors were then matched. Incorrect correspondences were rejected during geometric verification, and a homography was estimated using the remaining correspondences.

The homography provides a projective transformation between the two image planes:

$$x' = Hx$$

where H represents the 3×3 homography matrix.

The estimated transformation was subsequently used to warp the image into the panorama coordinate system. The warped image was then composited with the existing panorama.

Pairwise Results



Fig 3. ORB-based panorama obtained from Image 1 and Image 2



Fig 4. ORB-based panorama obtained from Image 2 and Image 3

The pairwise results demonstrate that ORB was able to identify sufficient correspondences to align the overlapping image regions.

Final Three-Image Panorama



Fig 5. Final panorama obtained using the ORB-based stitching pipeline

The final panorama successfully incorporates the three images and provides a wider field of view. However, a portion of the right side of Image 3 is not visible in the final output. This failure is discussed under **Limitations**.

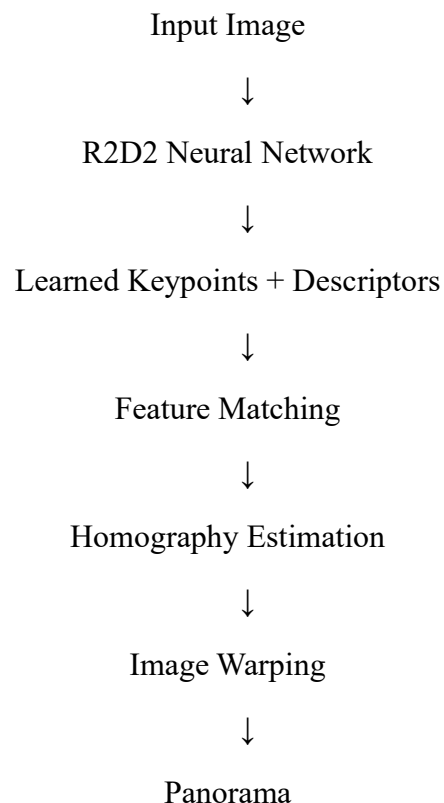
Deep Learning Method

For the deep-learning-based approach, **R2D2 (Repeatable and Reliable Detector and Descriptor)** was used. R2D2 was proposed by Revaud et al. [4] as a learned local feature detector and descriptor.

Unlike SIFT, ORB and SURF, whose feature representations are based on hand-designed algorithms, R2D2 learns feature representations using a neural network.

The R2D2 approach jointly considers the repeatability and reliability of local features. The objective is to identify image locations that can be detected consistently and provide reliable descriptors for matching.

The resulting pipeline can be represented as:



R2D2 Keypoints



Fig 6. Keypoints detected using R2D2

The detected R2D2 keypoints are concentrated around visually informative regions of the scene. Architectural structures containing corners, edges and local texture provide useful locations for feature matching.

R2D2 Feature Matching

Feature matching was performed for the three image pairs.



Fig 7. R2D2 feature matches between Image 1 and Image 2.



Fig 8. R2D2 feature matches between Image 1 and Image 3.

R2D2 Matching: Image 2 - Image 3



Fig 9. R2D2 feature matches between Image 2 and Image 3

The Image 1–Image 3 experiment is particularly useful because these images are not adjacent in the acquisition sequence. It provides an additional observation of the ability of learned descriptors to identify corresponding structures across a larger viewpoint difference.

R2D2 Stitching

The R2D2 features were subsequently used within the same geometric stitching framework.

R2D2 Panorama: Image 1 + Image 2



Fig 10. R2D2-based panorama for Image 1 and Image 2

R2D2 Panorama: Image 2 + Image 3

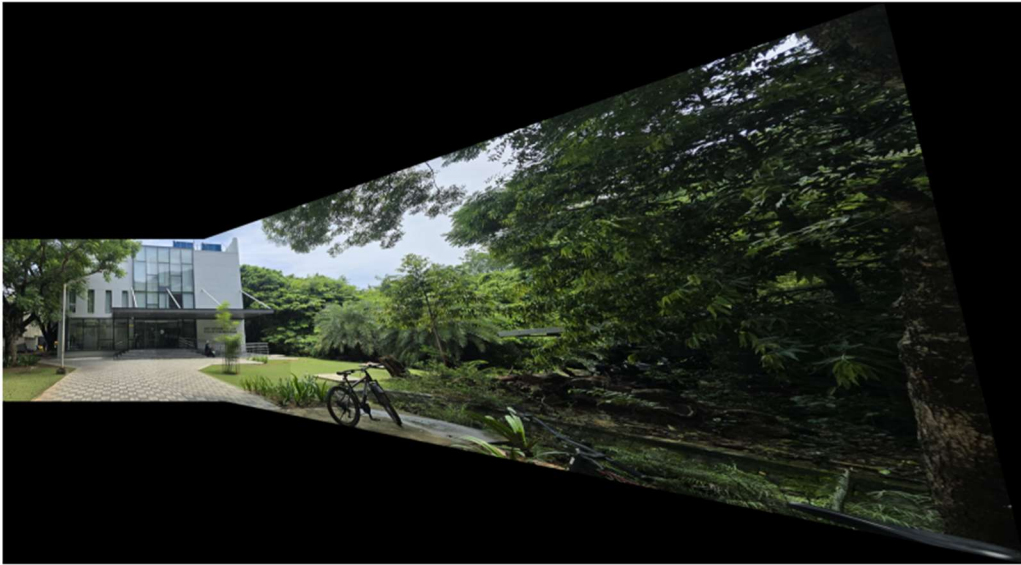


Fig 11. R2D2-based panorama for Image 2 and Image 3



Fig 12. Final three-image panorama generated using R2D2

The important observation is that changing the feature detector and descriptor does not change the fundamental geometric stitching process. Both ORB and R2D2 provide feature correspondences, after which homography estimation, warping and compositing are performed.

Classical vs Deep Learning

The main difference between the two approaches is the manner in which local image features are obtained.

Classical methods such as ORB use explicitly designed image-processing operations to detect and describe local structures. R2D2 instead learns feature representations from training data using a neural network.

Aspect	ORB	R2D2
Feature representation	Hand-crafted	Learned
Detector	FAST-based	Neural network
Descriptor	Rotated BRIEF	Learned descriptor
Descriptor type	Binary	Learned feature vector
Training	Not required	Required during model development
Inference	Lightweight	Neural-network inference
Matching	Hamming distance	Descriptor similarity
Computational requirement	Low	Higher
Adaptability	Limited by predefined design	Learned representation
Implementation complexity	Lower	Higher

ORB has an important practical advantage because feature extraction and matching can be performed with relatively low computational requirements. This makes it attractive for applications where processing speed and resource usage are important.

R2D2 requires neural-network inference and therefore introduces additional computational requirements. However, learned features have the potential to provide more adaptable representations for challenging visual conditions.

For this particular experiment, both approaches were able to establish correspondences and produce panorama results. Therefore, the quality of the final panorama depends not only on the

feature detector and descriptor but also on the quality of the matches, homography estimation and image compositing.

Limitations

Several limitations were observed during the experiment.

1. Incomplete Third Image

The most noticeable limitation was the incomplete visibility of Image 3 in the final ORB panorama. Although the three images were included, approximately a portion of the right side of the third image was outside the visible panorama.

This can occur when the warped image extends beyond the allocated panorama canvas. Therefore, correct calculation of the transformed image corners, panorama dimensions and translation offsets is important.

2. Sequential Homography Error

The panorama was constructed sequentially. Consequently, errors in the Image 1–Image 2 transformation can influence the position of Image 3 when the subsequent transformation is applied. Small geometric errors can accumulate toward the end of the panorama.

3. Limited Number of Images

Only three images from one scene were used. Therefore, the results do not represent all possible construction environments or camera configurations.

4. Camera Motion

The images were captured using a handheld smartphone. Small translation or rotation differences can affect the quality of the estimated homography.

5. Homography Assumption

The stitching pipeline uses a projective homography. This is suitable when the scene is approximately planar or when the camera undergoes mainly rotational motion. Significant three-dimensional depth differences can cause geometric inconsistencies.

6. Exposure Variation

Although the exposure compensation was constant at 0.0 EV, the shutter speed differed across the three images. Consequently, slight differences in brightness may occur between images.

7. No Advanced Seam Optimization

The implementation focuses on feature matching, homography estimation, warping and compositing. More advanced stitching systems can additionally use exposure compensation, seam optimization and multiband blending.

Conclusion

This assignment implemented an image stitching pipeline using both classical and deep-learning-based local feature methods.

SIFT, ORB and SURF were considered as classical feature-based approaches. Among them, ORB was selected for the final classical stitching pipeline because of its computational efficiency and compact binary descriptor.

The ORB pipeline successfully established correspondences between the overlapping images. In the Image 1–Image 2 pair, 105 matches were obtained, with 80 inliers after geometric verification, corresponding to an inlier ratio of 76.19%.

The deep-learning approach used R2D2 to generate learned local features and descriptors. These features were then incorporated into the same homography-based stitching framework.

The experiment demonstrates that the feature extraction method is only one component of a complete image stitching system. Correct matching, geometric verification, homography estimation, image warping and panorama canvas construction are all necessary for obtaining a successful result.

The incomplete right side of the third image in the ORB panorama highlights an important limitation of the current implementation. The issue can arise from accumulated transformation errors and panorama canvas/coordinate handling rather than simply from insufficient feature detection.

Overall, the experiment demonstrates the trade-off between classical and learned feature methods. ORB provides a simple, efficient and lightweight solution, while R2D2 provides a learned feature representation at the cost of greater computational and implementation complexity.

References

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GitHub Link: https://github.com/vasantharan/CV5570_Assignment_1 **

**Repo will be made public after submission deadline

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